

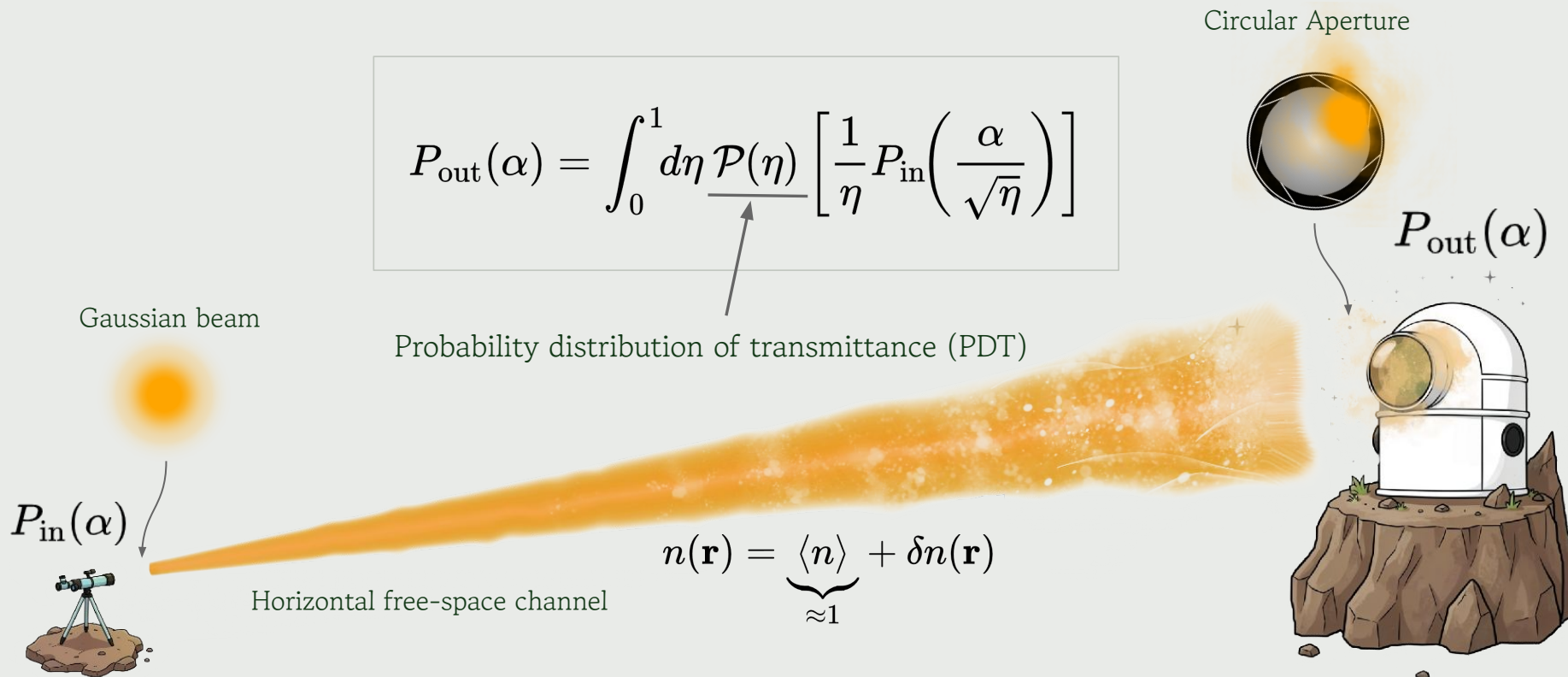


Bogolyubov Institute for Theoretical Physics
National Academy of Sciences of Ukraine

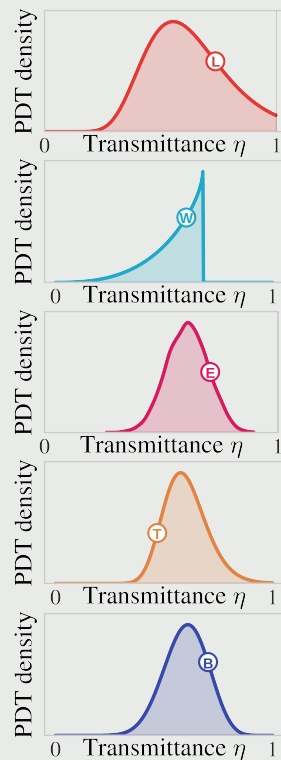
Statistical Models and Temporal Coherence of **Quantum Light** in the **Turbulent Atmosphere**

Mykyta Klen

Free-space quantum channels



Motivation



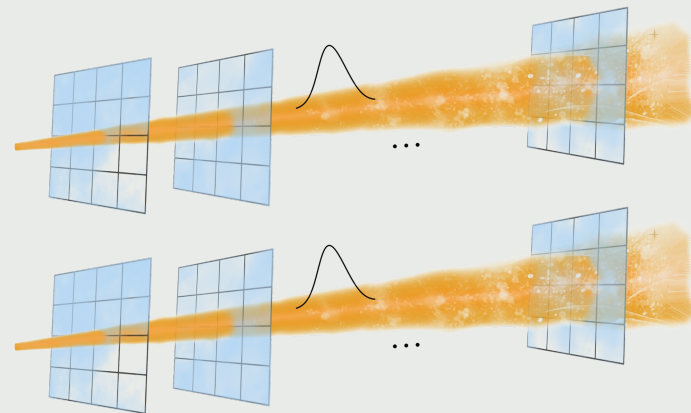
Truncated
log-normal model

Beam-wandering
model

Elliptical beam
model

Total-probability
model

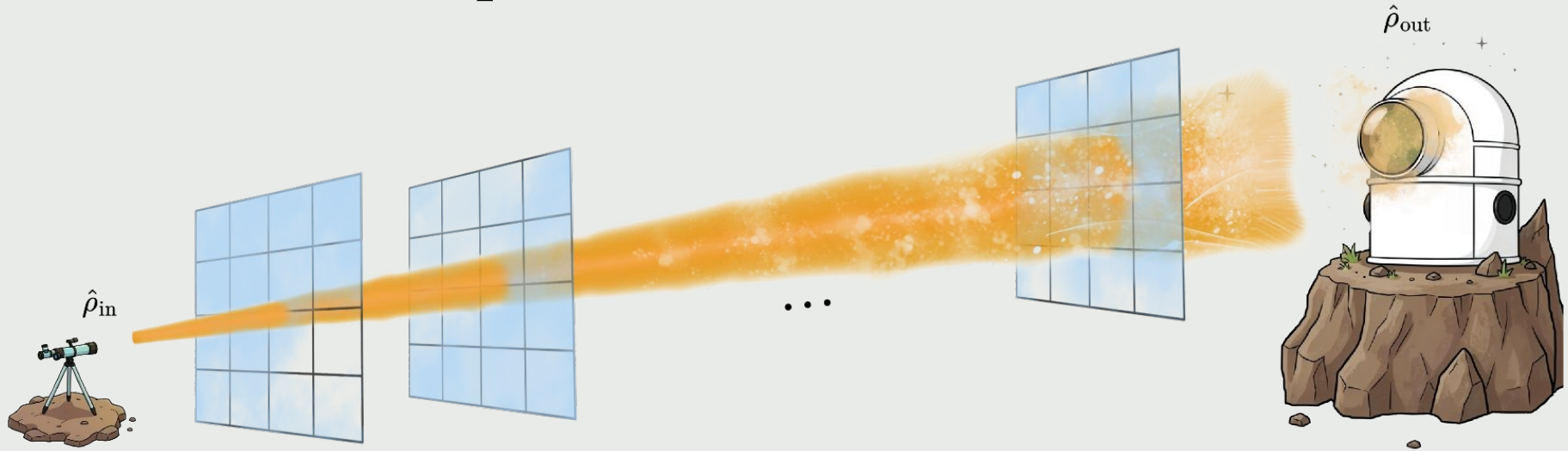
Beta-distribution
model



Ensemble of single pulses

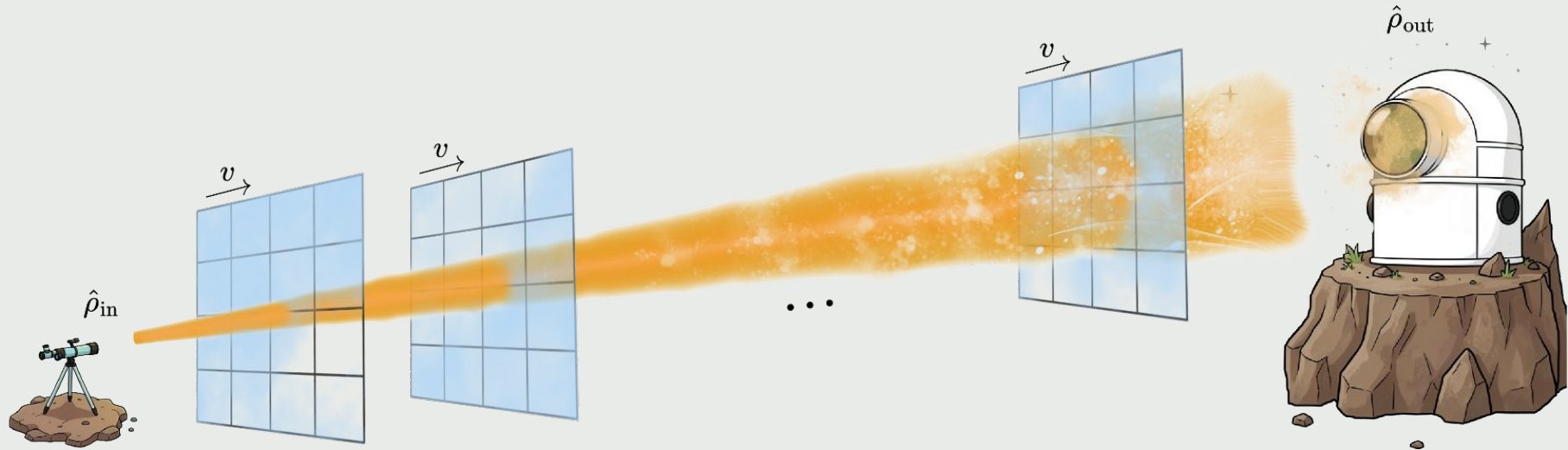
Methodology: Phase-screens numerical simulation

$$n(\mathbf{r}) = \underbrace{\langle n \rangle}_{\approx 1} + \delta n(\mathbf{r})$$



Time correlations: Frozen-turbulence hypothesis

$$n(\mathbf{r}, t) = n(\mathbf{r} - \mathbf{v} t, 0)$$



TOC

1. Background
2. Validation of analytical models
3. Statistical properties of the beam shape parameters
4. Moment matching for the Circular-beam model
5. Time correlations in Atmospheric Quantum Channels
6. Applications
 - 6.1 Gaussian entanglement between pulses
 - 6.2 Discrete-variable entanglement between pulses
 - 6.3 Threshold-based selection for nonclassical states

Klen & Semenov, Phys. Rev. A 108, 033718 (2023)

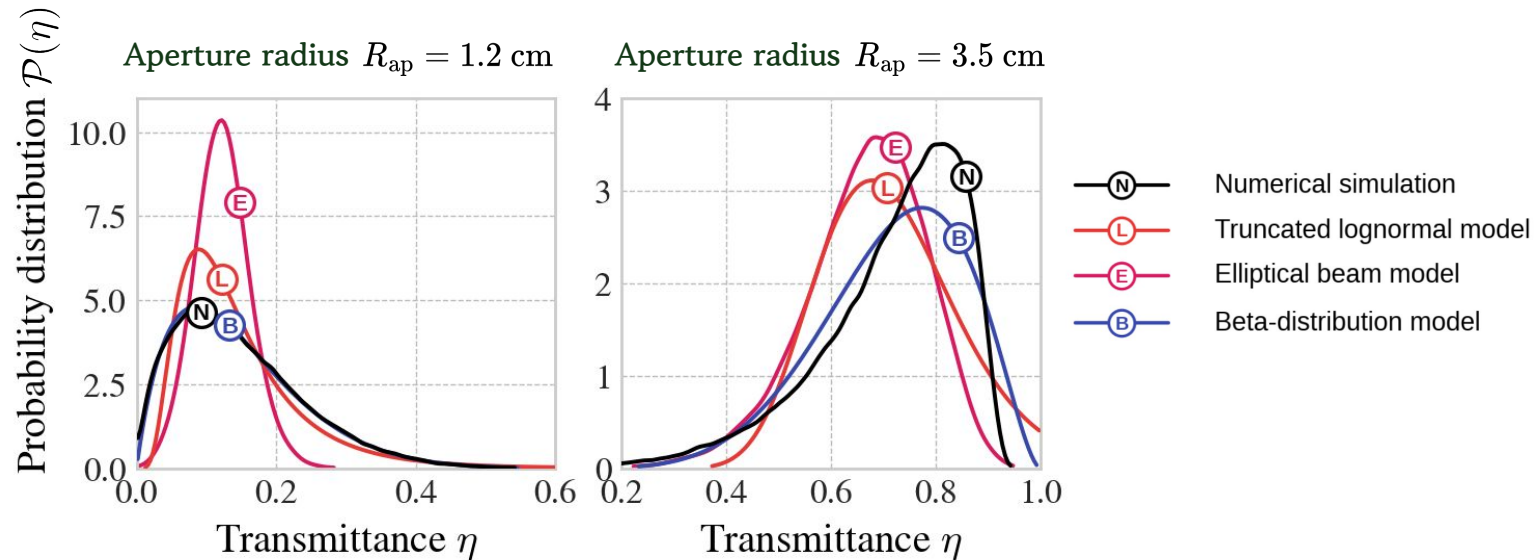
Pechonkin, Klen & Semenov
Phys. Rev. A 112 063716 (2025)

Klen et al., Phys. Rev. A 109, 033712 (2024)

Validation of analytical models

Results of PDT simulations

Klen & Semenov, Phys. Rev. A 108, 033718 (2023)



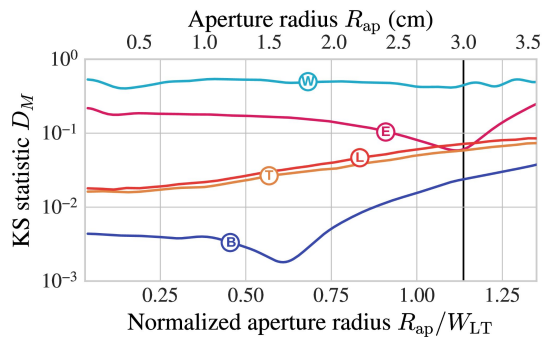
Validation of analytical models

Klen & Semenov, Phys. Rev. A 108, 033718 (2023)

$$\sigma_R^2 = 1.23 C_n^2 \left(\frac{2\pi}{\lambda} \right)^{7/6} z_{\text{ap}}^{11/6}$$

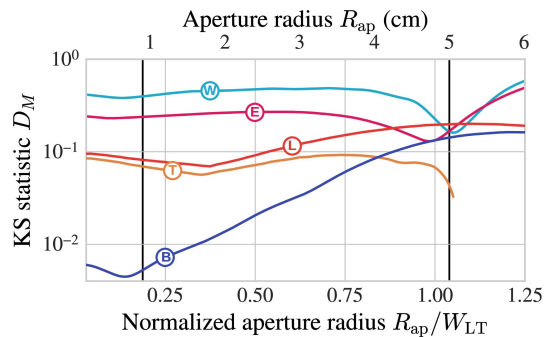
Cumulative distribution functions

$$D_M = \sup_{\eta} |F_M(\eta) - F(\eta)|$$



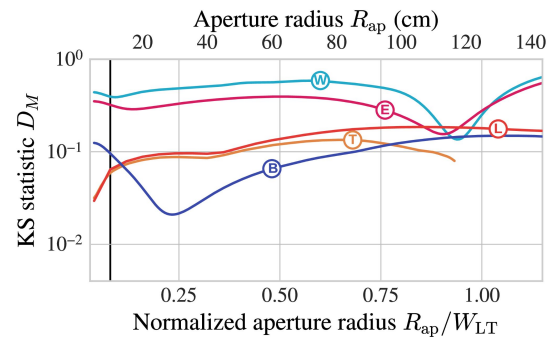
Weak turbulence

1 km, $\sigma_R^2 = 0.2$



Moderate turbulence

1.6 km, $\sigma_R^2 = 1.5$



Strong turbulence

50 km, $\sigma_R^2 = 33.3$

— (N) —

Numerical simulation

— (L) —

Truncated lognormal model

— (W) —

Beam wandering model

— (E) —

Elliptical beam model

— (T) —

Total probability model

— (B) —

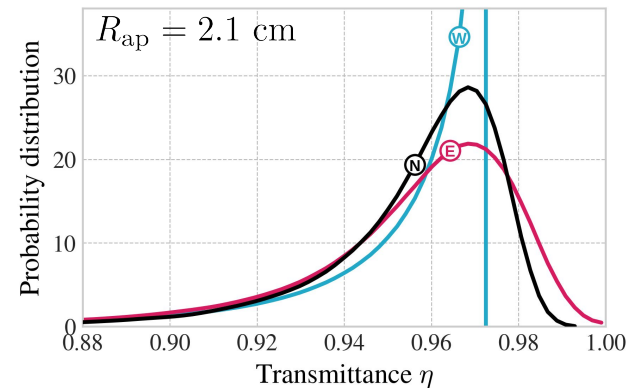
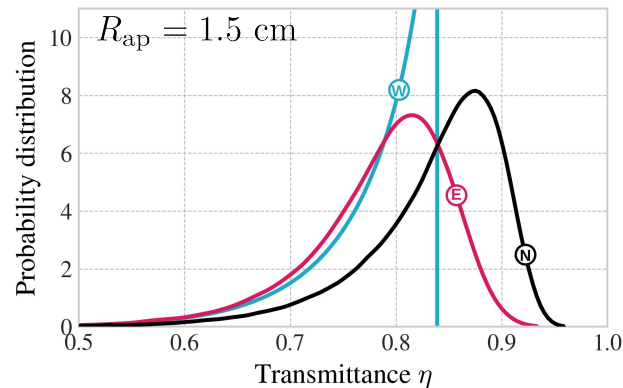
Beta-distribution model

Moment matching for the Circular-beam model

Misspecification bias in physics-based models

Pechonkin, Klen & Semenov, Phys. Rev. A 112, 063716 (2025)

$$\mathcal{P}(\eta; \langle x_0^2 \rangle, \langle W^2 \rangle, \dots)$$



—(N)—

Numerical simulation

—(W)—

Beam wandering model

—(E)—

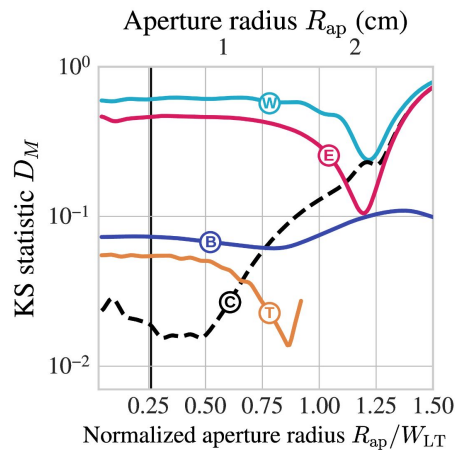
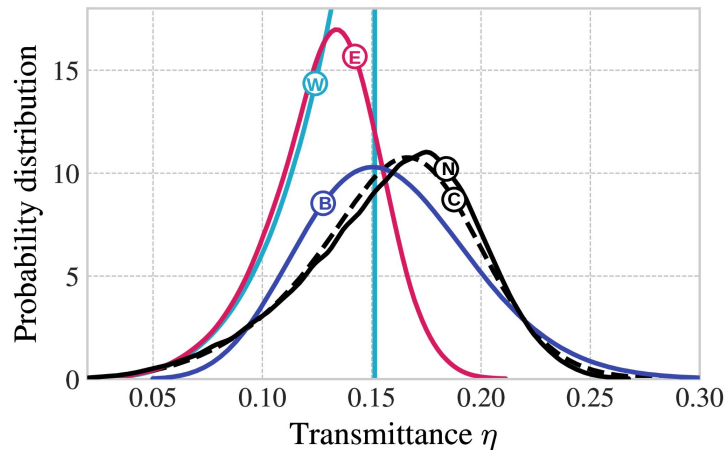
Elliptical beam model

Weak turbulence
1 km, $\sigma_R^2 = 0.2$

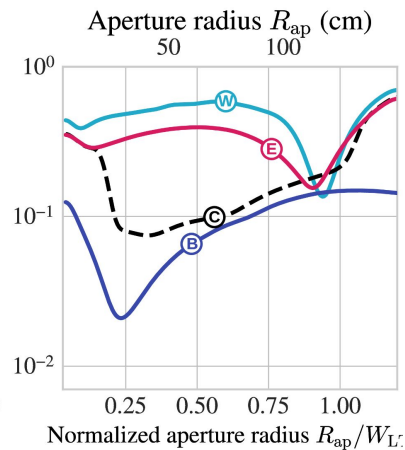
Moment matching for physics-based models

Pechonkin, Klen & Semenov, Phys. Rev. A 112, 063716 (2025)

$$\mathcal{P}(\eta; \langle x_0^2 \rangle, \langle W^2 \rangle, \dots) \longrightarrow \mathcal{P}(\eta; \langle \eta \rangle, \langle \eta^2 \rangle, \dots)$$



Weak turbulence



Strong turbulence

Time correlations in Atmospheric Quantum Channels

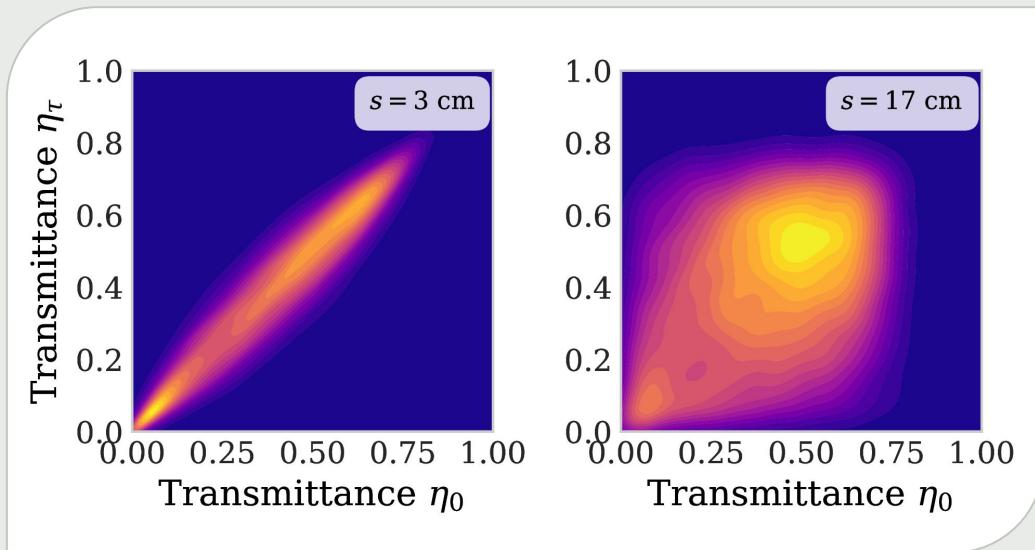
Time correlations: The Two-Time PDT

Klen et al., Phys. Rev. A 109, 033712 (2024)

$$s = v\tau$$

If $v = 10 \text{ m/s} \implies 1 \text{ cm} \equiv 1 \text{ ms}$

$$P_{\text{out}}(\alpha) = \int_{\Xi} d\boldsymbol{\eta} \mathcal{P}(\boldsymbol{\eta}) \left[\frac{1}{\eta_0 \eta_\tau} P_{\text{in}} \left(\frac{\alpha_0}{\sqrt{\eta_0}}, \frac{\alpha_\tau}{\sqrt{\eta_\tau}} \right) \right]$$



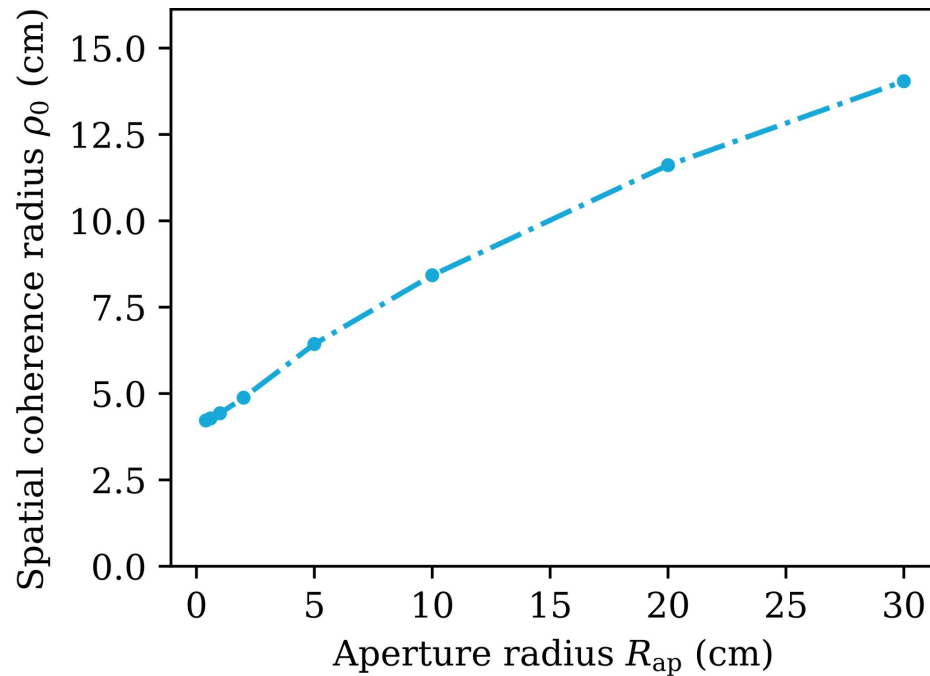
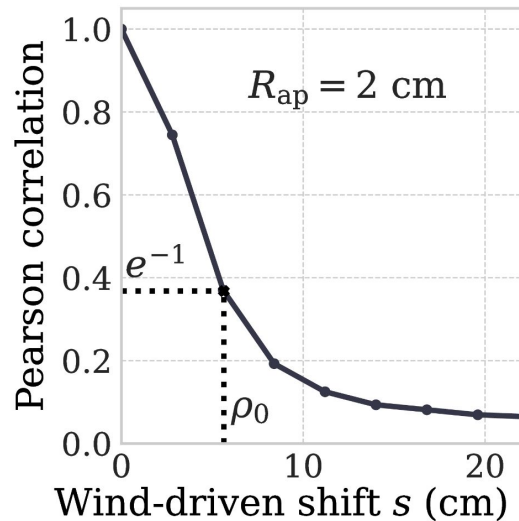
$$\boldsymbol{\eta} = (\eta_0 \quad \eta_\tau)$$
$$\Xi = [0, 1] \times [0, 1]$$

$$\mathcal{P}(\eta_0) = \int_0^1 d\eta_\tau \mathcal{P}(\eta_0, \eta_\tau)$$

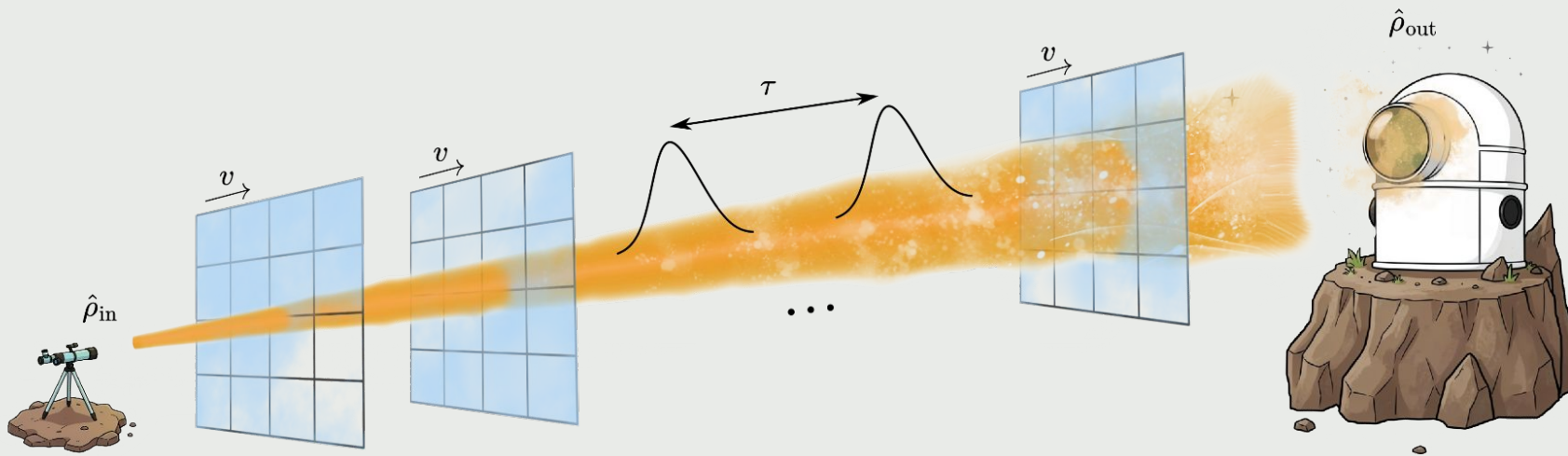
$$\sigma_R^2 = 5.5, L = 50\text{km}, R_{\text{ap}} = 20\text{cm}$$

Time correlations: Spatial Coherence Radius

Klen et al., Phys. Rev. A 109, 033712 (2024)



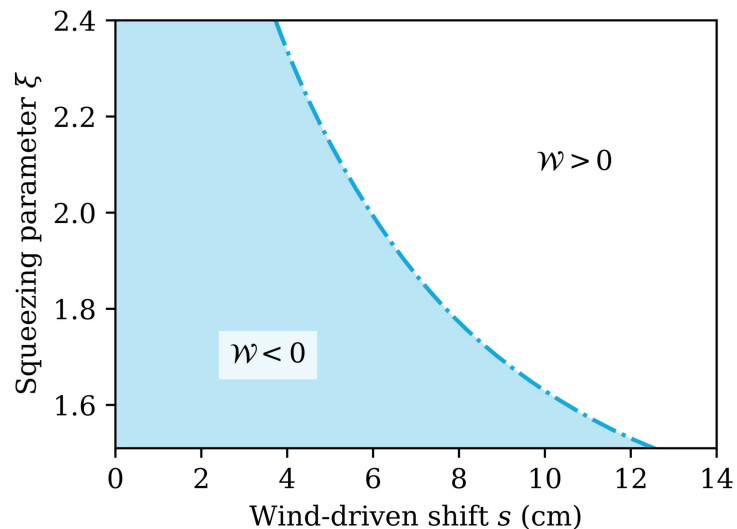
Applications



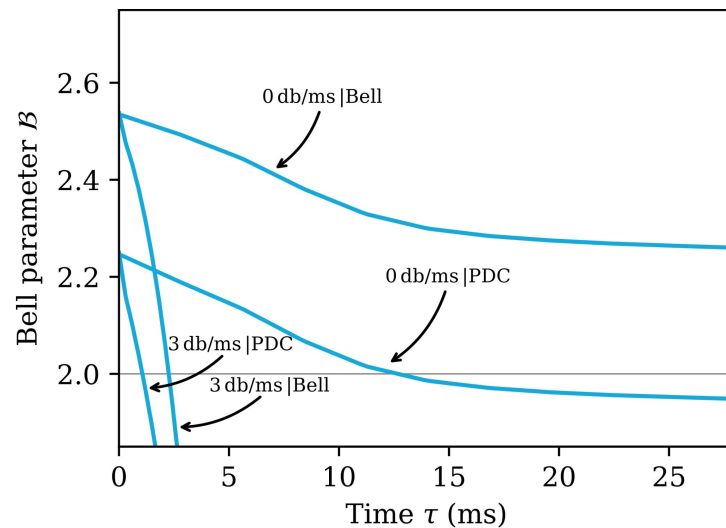
Entanglement

Klen et al., Phys. Rev. A 109, 033712 (2024)

The two-mode squeezed vacuum (TMSV) states



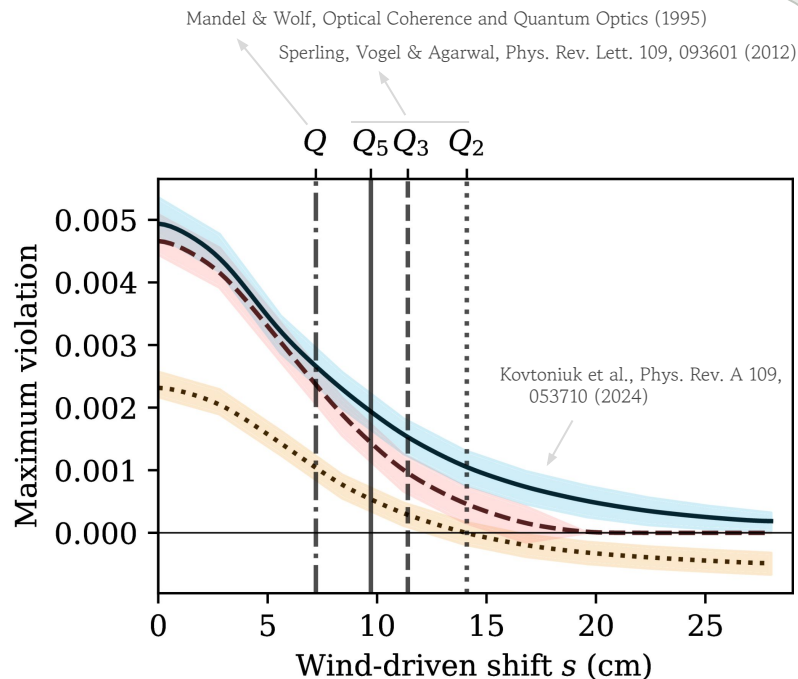
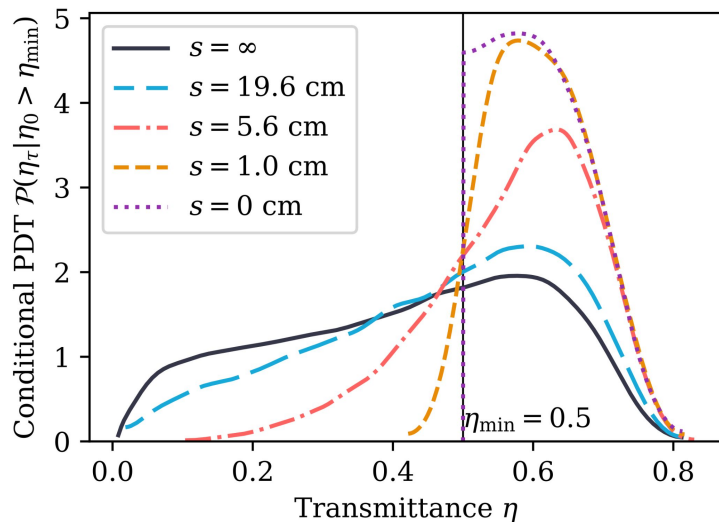
Bell & PDC states



Adaptive Selection Protocols

Klen et al., Phys. Rev. A 109, 033712 (2024)

Conditional PDT



* Amplitude-squeezed coherent states

Conclusions

Aperture size as primary control parameter

The ratio of receiver aperture to average beam size is the primary parameter shaping the transmittance distribution, overriding traditional heuristics based only on turbulence strength.

Practical Beta-distribution model

An empirical Beta distribution model provides the most accurate fit across all turbulence regimes because it naturally handles the transmittance range and captures skewness transitions

Elimination of misspecification bias

Traditional beam-shape based models suffer from the misspecification bias. The introduction of a transmittance-moment matching technique resolves this, significantly improving model accuracy of the physic based models.

Temporal correlations

The two-time PDT framework and coherence radius ρ_0 quantify correlations. ρ_0 scales almost linearly with aperture size.

Quantum robustness

Entanglement persists over millisecond timescales.

Discrete-variable entanglement is limited in practice by quantum memory losses rather than atmosphere alone.

Adaptive selection

Protocols using classical probes effectively leverage temporal correlations to enhance nonclassicality over pulse separations of tens of milliseconds.

Publications

- [1] **Klen, M., & Semenov, A. A. (2023)**. Numerical simulations of atmospheric quantum channels. Physical Review A, 108(3). <https://doi.org/10.1103/physreva.108.033718> (Q1)
- [2] **Klen, M., Vasylyev, D., Vogel, W., & Semenov, A. A. (2024)**. Time correlations in atmospheric quantum channels. Physical Review A, 109(3). <https://doi.org/10.1103/physreva.109.033712> (Q1)
- [3] **Pechonkin, I., Klen, M., & Semenov, A. A. (2025)**. Circular-beam approximation for quantum channels in a turbulent atmosphere. Physical Review A, 112(6). <https://doi.org/10.1103/pv7j-4zpf> (Q1)
- [4] **Semenov, A., Klen, M., & Pechonkin, I. (2025)**. Quantum optics in the turbulent atmosphere: fundamental issues and applications. Quantum Technologies for Defence and Security II (p. 38). SPIE. <https://doi.org/10.1117/12.3069599>
-

Conferences and scientific meetings

The results of the dissertation were presented at seminars of the Bogolyubov Institute for Theoretical Physics of the National Academy of Sciences of Ukraine, as well as at Ukrainian and international conferences, in particular:

1. M. Klen and A. A. Semenov, "Free-space quantum channels: Numerical simulations", US-Ukraine Quantum Forum 2023, August 28-31, 2023
 2. M. Klen, "Quantum light in atmospheric turbulence", 25th Symposium on photonics and optics SPO 2024, Kyiv, (Ukraine), November 8, 2024
 3. M. Klen, "Numerical simulations in free-space quantum channels", poster session at Quantum 2025: From Foundations of Quantum Mechanics to Quantum Information and Quantum Metrology & Sensing, Turin, (Italy), May 18-24, 2025
 4. A. Semenov, M. Klen, I. Pechonkin, "Quantum optics in the turbulent atmosphere: fundamental issues and applications": invited talk at the SPIE Sensors+Imaging Meeting. Quantum Technologies for Defense and Security II. Madrid, Spain. September 15-18, 2025.
-

Thank You!

Scientific advisor

Andrii Semenov

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Lukas Bulla

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Vadym Kovtoniuk

Illia Pechonkin

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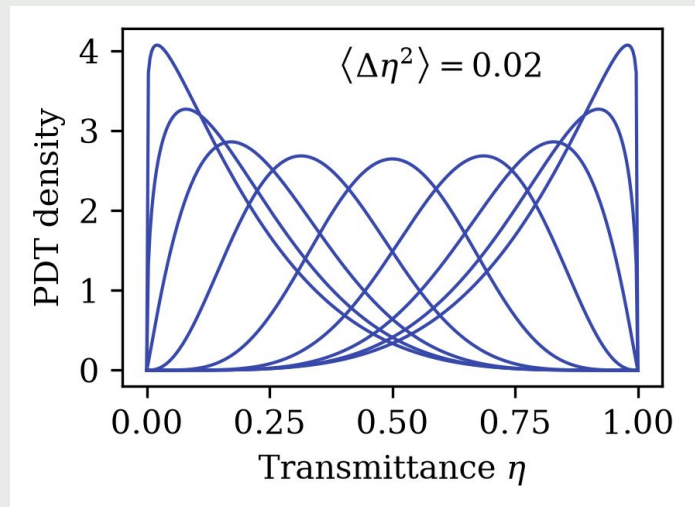
Simons Foundation

Research programs of the

NAS of Ukraine

Question 1

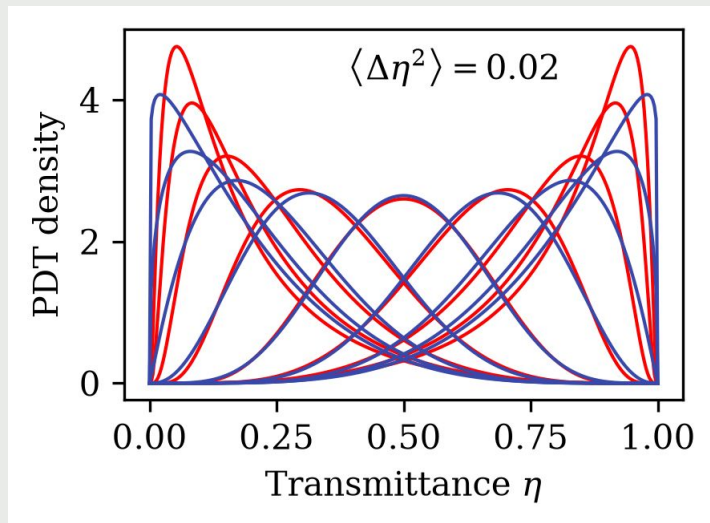
Can the author provide a physical interpretation explaining why the [Beta-distribution](#) reproduces atmospheric transmittance statistics so successfully over such a broad parameter range?



$\langle \eta \rangle \in (0.175, 0.2, 0.25, 0.35, 0.5, 0.65, 0.75, 0.8, 0.825)$

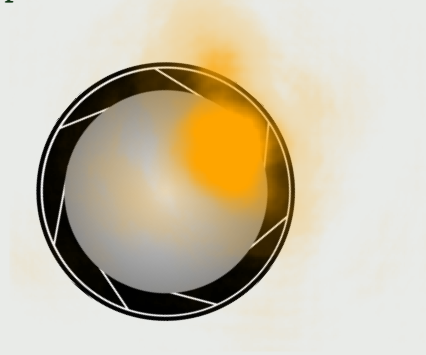
Question 2

Have alternative bounded distributions, such as [logistic-normal distributions](#) or other transformations of Gaussian random variables, been considered and compared with the Beta distribution model?



Question 3

To what extent does the author attribute the limitations of beam-shape-based models to turbulence-induced speckle formation rather than to smooth beam-shape distortions?



Question 4

Has the author performed a direct analysis of spatial intensity distributions to quantify the role of speckles in the observed model discrepancies?

Question 5

To what extent does the effective beam-size parameter retain a clear physical meaning after the Transmittance-Moment Matching procedure is applied?

Question 6

Is it possible to generalize the moment-matching approach to the elliptical-beam model, or does the increased number of parameters fundamentally limit such a generalization?

Question 7

How sensitive are the obtained temporal correlation times and the introduced coherence radius to deviations from the frozen-turbulence approximation?

Question 8

In the hypothetical absence of wind, what physical mechanism would determine the characteristic decorrelation time of atmospheric transmittance?

Question 9

Would it be more appropriate to interpret the introduced quantity as a correlation length of transmittance fluctuations rather than as a coherence radius in the traditional optical sense?

Question 10

Which of the results obtained in the dissertation does the author consider to be genuinely specific to quantum communication, and which results should be regarded as more general contributions to statistical atmospheric optics?

Question 1

We can see that the choice of references does not include results from very recent years, thus losing a fair amount of useful results, given the great progress of the sector. Including some of the latter ones would be helpful.

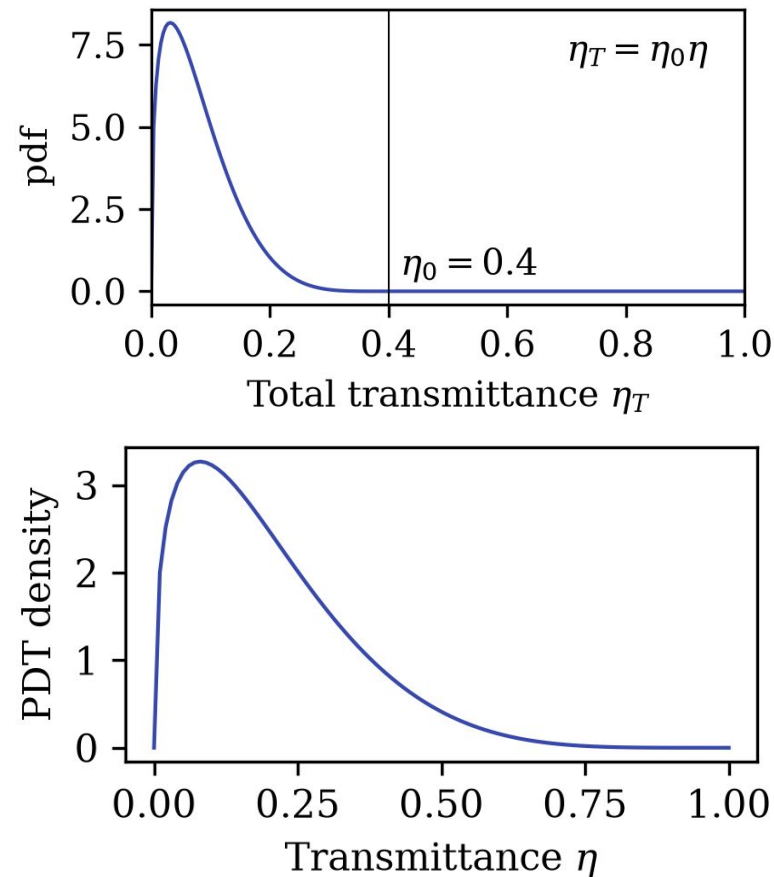
Question 2

We note that on the development of the numerical analysis, the setting and numerosity of the phase screens are related to the intrinsic atmospheric scale. In this context, a different approaches uses instead the reference to the observation-plane region of interest, and for the spatial frequency

limits, the detector scale. This is discussed for example in the manual by Schmidt and detailed in its Chap. 7 . A discussion on the relation of this more phenomenological viewpoint would be useful. In this regards, also the analysis of the angle-of-arrival is relevant, as it conditions the optical design of the receiver and of potential turbulence correctors.

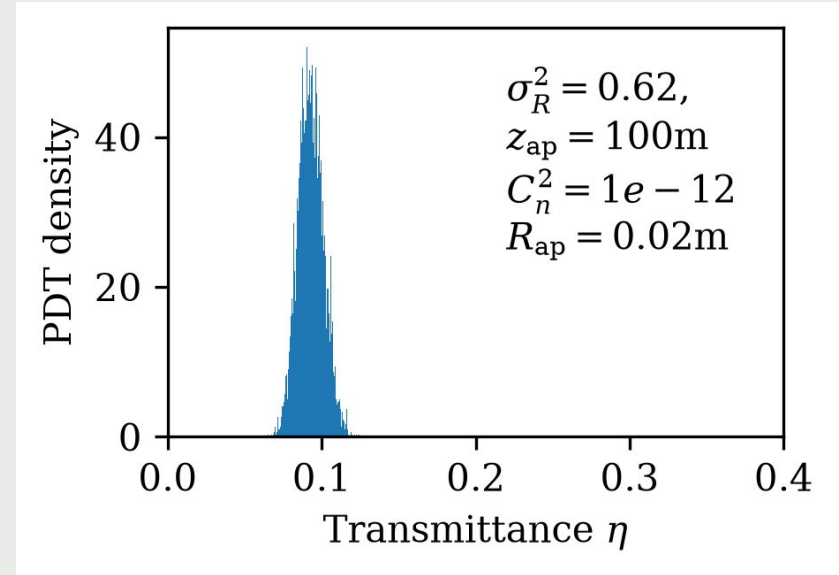
Question 3

We note that the validation of the analytical models are realized by means of numerical simulations, with three different Rytov variances. Now, the probability distribution of transmittance (PDT) and the beam-centroid spread have been assessed for many real channels and may provide a valuable reference for the predictions. This from the pure wandering case of short tracks to the extreme cases of the Canary link, beside the satellite cases (JAXA, ESA and then CAS). Only to cite our own results, results on the different lengths are reported in literature. A discussion on these references may be useful too.



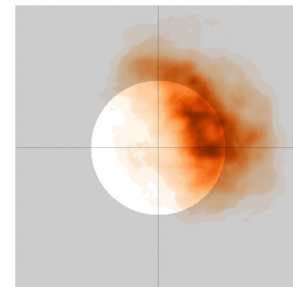
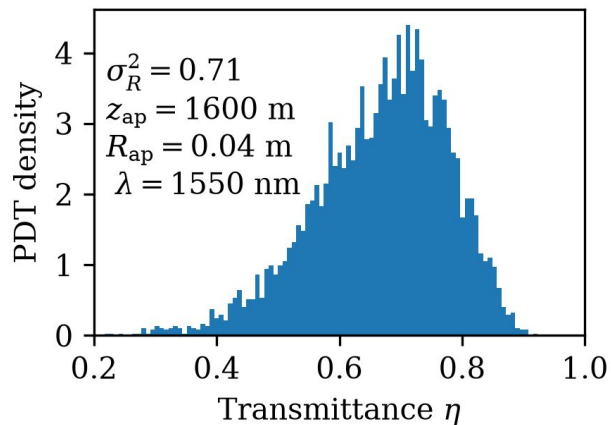
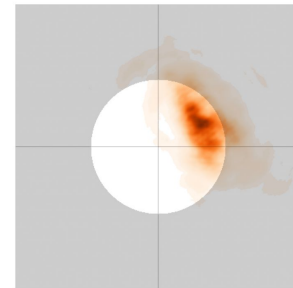
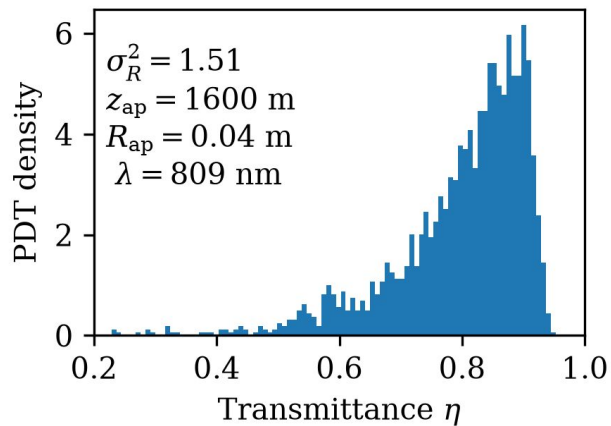
Question 4

We note that in the selection of the channel lengths to be analysed, the very short range (interbuilding 100-500 m) and the metropolitan on (3-10 km) were not considered. As they correspond to several applications, a discussion would be useful.



Question 5

The selection of the wavelength for the quantum states is also relevant. Many results have shown advantages in reduced background light and larger Fried parameter moving toward C-band. Indeed the almost linear scaling of this latter crucial parameter supports the adoption of 1550 nm light. This also allows the use of the telecom technology, that is both very advanced and cheaper than for NIR. Also on the wavelength dependence of the results, a discussion would be useful.



Question 6

By taking the case of the example of the writing of a flying qubit on a quantum memory in sec. 6.2, the scheme generally considered in the Thesis is of free-space detection. While it's possible to interact with the memory active medium with a multimode beam, the subsequent reading pulse cannot reproduce the turbulence-induced spatial spectrum of the writing, causing a significant drop in fidelity. For this reason as well as for more technical ones as the necessary ultra narrow spectral filtering, the memory is usually made to interact with a single spatial mode, either guided or free space. In addition, C-band light usually foresees the quantum measurement and the detection to be realised with single-mode fiber systems.

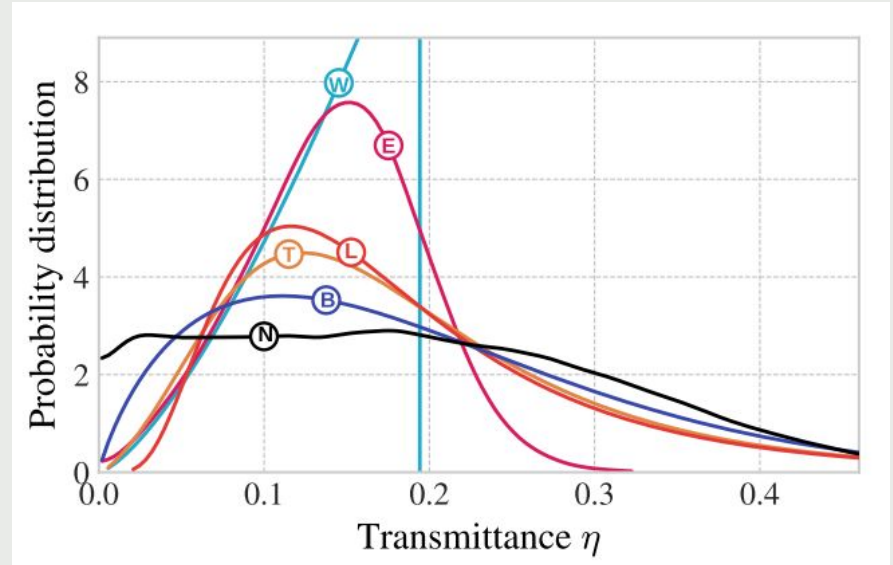
This aspect is imposing a more advanced criteria for the entire transmissivity assessment. Indeed the relation changes from Eq. 5.3 to one in which the integrand function contains the single mode function as $u_t(\mathbf{r}, z)\psi^*(\mathbf{r})$ and not just $|u_t(\mathbf{r}, z)|^2$. This aspect also relates on the mitigation of turbulence with active/adaptive means, very common for example in the classical coherent detection in satellite communications. This is very relevant in the situation of quantum channels, in which the losses cannot be mitigated by increasing the transmitted power. Some analysis are available, including from ours, and a discussion on this aspect would be very useful.

Question 7

Finally we note that the result about the persistence of quantum entanglement and nonclassicality in general over millisecond to tens-of-milliseconds timescales in turbulent atmospheric channels is a very useful notion for advanced experimental scheme. An analysis of potentialities may be included. As example, it may be used in the study of bringing the Colella-Overhauser-Werner (COW) interference test in the space context for assessing general relativity effect on photonic quantum states with a required relative precision in the phase estimate of the order of 10^{-8} as described in the references.

Question 1

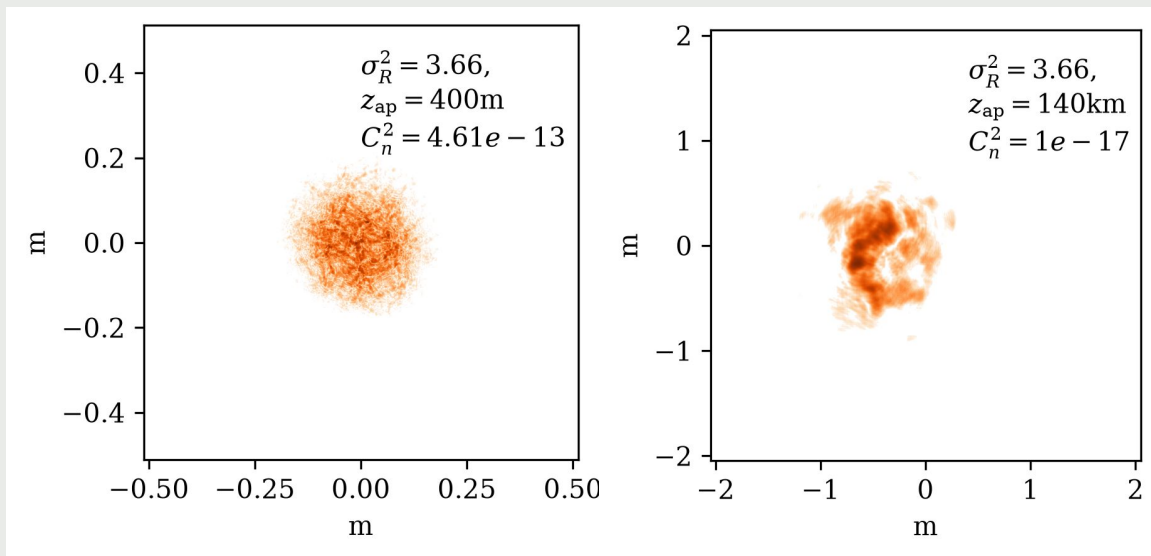
Can we have some physical intuition on the plateau of the numerical PDT observed for a focused beam in intermediate turbulence in Fig. 3.8?



Question 2

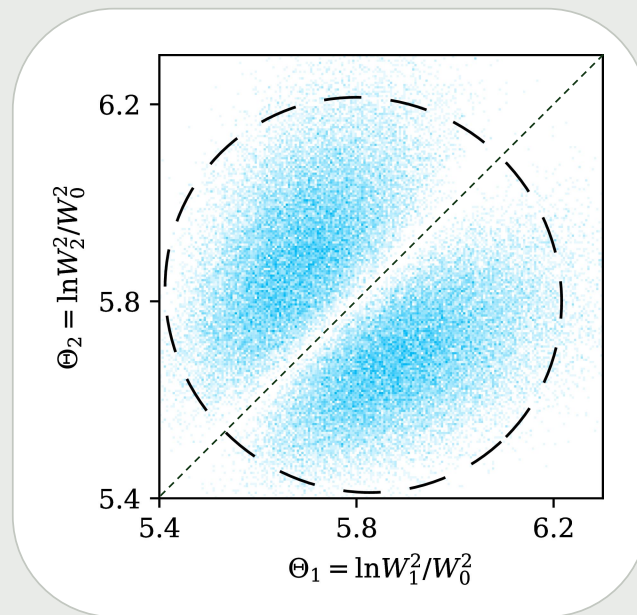
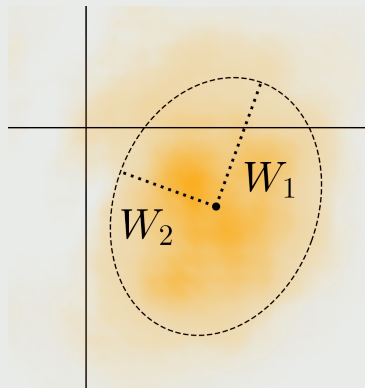
Given the strong dependence of the PDT on the aperture size, diffraction-induced beam broadening appears to play an important role. The channels investigated in the thesis are classified through their Rytov variance, which combines diffraction effects (through the propagation distance z_{ap}) and turbulence effects (through the turbulence structure constant C_n^2). It would therefore be interesting to compare two channels with the same Rytov variance but different values of z_{ap} and C_n^2 . Would one expect the resulting PDTs to be essentially identical, suggesting that the Rytov variance is the dominant parameter,

or could significant differences arise from the separate contributions of diffraction and turbulence? In other words, is the behavior of the PDT primarily diffraction-dominated, turbulence dominated, or determined by the interplay between the two effects?



Question 3

How can I physically interpret the suppression of diagonal points in Fig. 4.4?



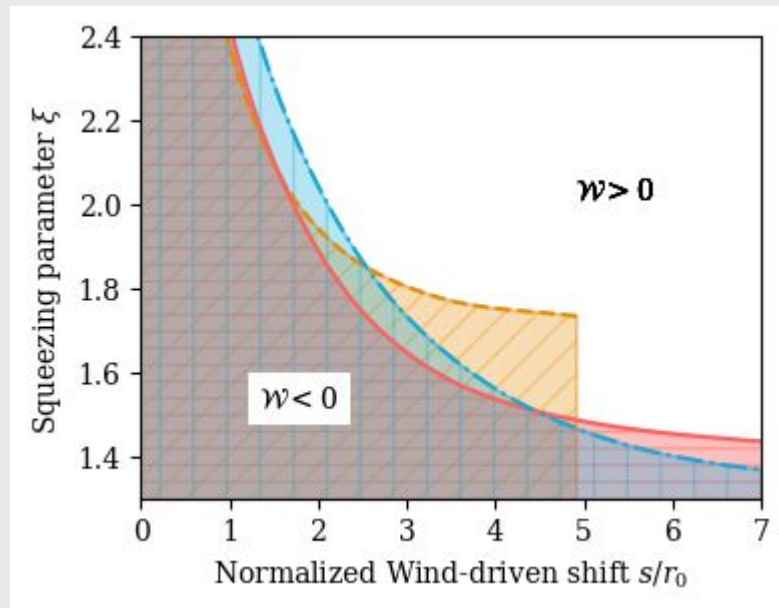
$$\mathbf{S} = \begin{pmatrix} S_{xx} & S_{xy} \\ S_{xy} & S_{yy} \end{pmatrix}, \quad W_{\pm}^2 = \frac{1}{2} \left(S_{xx} + S_{yy} \pm \underbrace{\sqrt{(S_{xx} - S_{yy})^2 + 4S_{xy}^2}}_{R=\sqrt{X^2+Y^2}} \right)$$

Question 4

In the thesis, a constant wind velocity for all phase screens is used to derive the two-time PDT. This simplification is justified by the need to maintain the model "identifiable and tractable". Another common approach to account for inhomogeneities, while keeping the model "identifiable and tractable" is to draw the screen velocities from a Gaussian distribution with given mean and variance. Could one expect significant changes in the results from moving to a less homogeneous model?

Question 5

In Chapter 6, the temporal properties of the channel are presented in terms of absolute wind-induced shifts (cm). Since the relevant scales are ultimately determined by turbulence and beam propagation, would it be more informative to express these results in terms of characteristic turbulence-induced length scales, such as the Fried parameter r_0 or the long-term beam radius W_{LT} ? Would such a normalization reveal a more universal behavior across different channel conditions?



Question 6

The characteristic pulse rate for adaptive free-space quantum communication is determined by the ratio v/ρ_0 , where ρ_0 is the spatial coherence radius and v is the transverse wind speed. This quantity appears conceptually similar to the Greenwood frequency, which scales as v/r_0 and sets the required correction-loop bandwidth in adaptive optics. In both cases, the underlying requirement is that the channel remains sufficiently correlated between its characterization and the subsequent action, whether transmitting a quantum state or applying a wavefront correction. Could the author comment on the relationship between these two quantities?

Question 7

How was the threshold used in the study of the adaptive protocol in Section 7.3 chosen? I would expect a trade-off between exploiting only the highest-transmittance channel realizations and maintaining a sufficiently large number of transmission opportunities. Can the author comment on this interplay? More specifically, is there an optimal threshold that balances the quality of the transmitted quantum states against the resulting communication rate?

Question 1

The Author uses the Rytov parameter to characterise the strength of turbulence of the atmospheric channels. However, the thesis would benefit from an introductory discussion about the real meteorological quantities, such as temperature or pressure, that correspond to the values of the Rytov parameter being discussed. Especially because the Authors devoted a lot of effort towards discussing the physically realistic parameters.

Question 2

This work relies extensively on the Taylor frozen hypothesis. While it is widely accepted for the weak turbulence, recent research on the strong turbulence in the atmospheric surface layer (which is up to 100-150 meters above the ground), this hypothesis does not always work [see Physics of Fluids 34, 076602 (2022), DOI: <https://doi.org/10.1063/5.0097729>]. Therefore, the limitations of the Taylor frozen hypothesis for quantum communications within the atmospheric surface layer should be addressed more carefully.

Question 3

The purpose of Figure 1.1 is not clear. Is it an illustrative example of a random function? Even if it is so, it was not explained how it was calculated.

Question 4

This comment concerns Figures 3.1-3.3.

(a) Figures 3.1 and 3.2 need labels like (a) and (b) or legends so that the reader can easily distinguish between the collimated and focused beam cases.

(b) In Figures 3.1-3.2, the dependencies for the collimated and focused beams are almost identical, and the deviations from the Gaussian law are not clearly seen. It would be more beneficial to superimpose two dependencies and to provide a zoomed area to highlight the most interesting areas. The latter is also true for Figure 3.3.

(c) The quality of the data presentation is low. For example, Figure 3.3 seems to correspond to Figure 9 of the paper [I]. While in the paper the figure looks clearly as a histogram, in the one in the thesis, the separate columns cannot be distinguished, and the histogram-like nature of the figure is lost.

Question 5

On page 53, line 9 from the bottom, there is a typo in the reference. I do not understand what “[^@bachmann2025]” stands for.

Question 6

The term “spatial frequency” is used several times throughout the thesis (pages 38,51) instead of the more conventional term “wave number”.